CHAPTER 3

CSS(CASCADING STYLE SHEETS)

Quiz one ..Create the ff page.Boxes represent a div element.

Css Example

This is the first pagaraph [Link one with title home]

About US

This is the second pagaraph [Link Two with title about]

This is the third pagaraph [Link Three with title contact Us]

What is CSS?

- Every web page is composed of HTML (Hypertext Markup Language) code that describes the content
- Example:

  <p>Animportant<font

  color=“#FFFF00”>paragraph.</p>
  - Displays:
    - An important paragraph.
    - Repetitive and hard to maintain.

What is CSS?

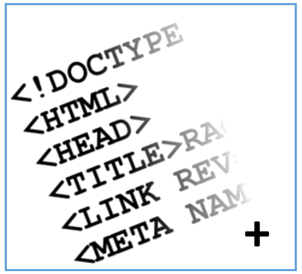
- Layers of a web page:
  - **Content:**
    - Text, images, animation, video, etc.
  - **Presentation:**
    - How the content will appear to a human through a web browser, text reader, etc.
  - **Behavior:**
    - Real-time user interaction with the page: validation, sorting, drag-n-drop etc.

Style Sheet Languages

- Style sheet languages are used to describe the presentation of structured documents, like HTML, XML and other markup languages.

What is CSS?

- Cascading Style Sheets
- Contains the rules for the presentation of HTML



+

=

- CSS was introduced to keep the presentation information separate from HTML markup (content).
- CSS separates the presentation from the content.

What are Cascading Style Sheets?

- Cascading Style Sheets (CSS) are *rules*. Each rule consists of a *selector* and a *declaration* (which, in turn, is made up of a *property* and a *value*).

- They were established by the World Wide Web Consortium (W3C). CSS rules control the look (*Style*) of web pages or XML files by providing central locations (*Sheets*) where HTML or XML tags are interpreted by the browser.

Why CSS?

- Flexible, can be applied in several ways.
- Easy to maintain.
- Accessibility to different users with different devices.
- CSS caching = less bandwidth + fast loading

Why use Style Sheets?

Every keystroke counts!
Smaller files load more quickly
Save disk space

For Example

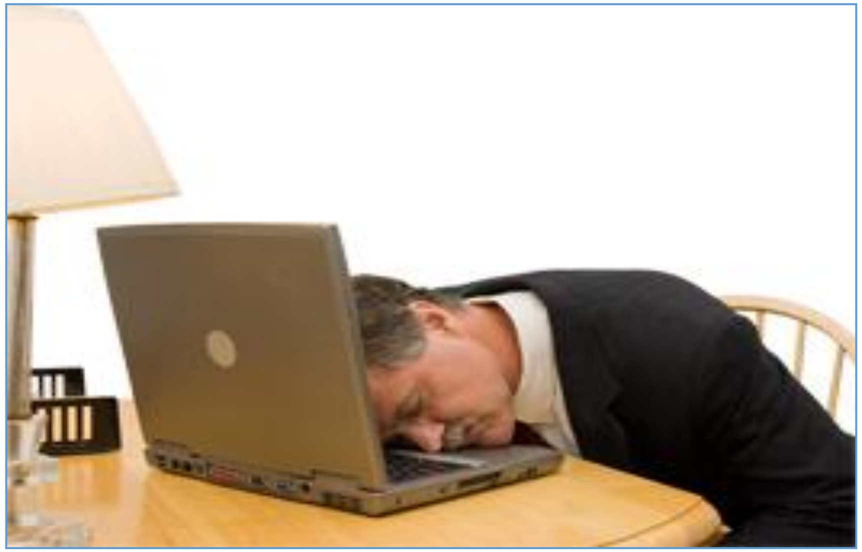
Given a page *webreg.html*

Original: 27.2K
Embedded Style Sheet: 26.2K
External Style Sheet: 25.6K

- separate structure from appearance
- create consistent appearance
- ease of maintenance
- increase accessibility
- apply additional effects
- reduce use of non-standard tags
- reduce web page file size

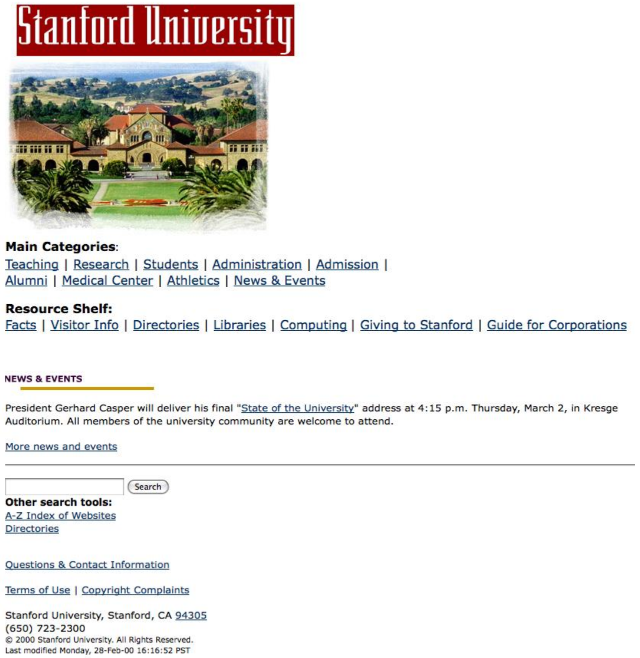
What is CSS?
> Before CSS

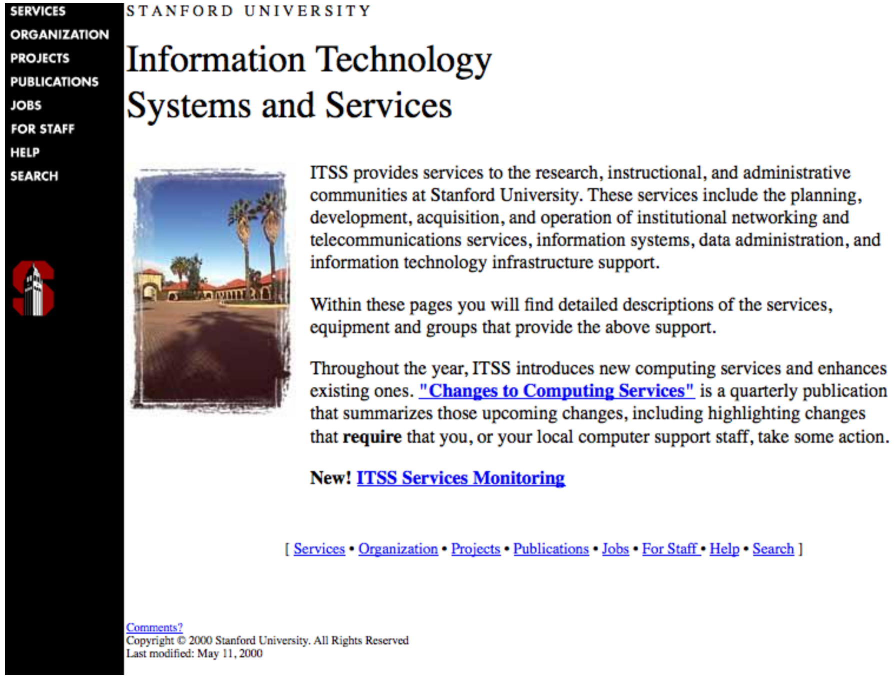
- Initially designers used presentation tags like (FONT, B, BR, TABLE ETC.) and spacer GIFs to control the design of web pages.
- Any **modification** in the design of websites was a very **difficult** and **boring** task, as it involves **manually editing** every HTML page.
- Providing support for multiple browsers was a difficult task.



Brief history… 1997-2001

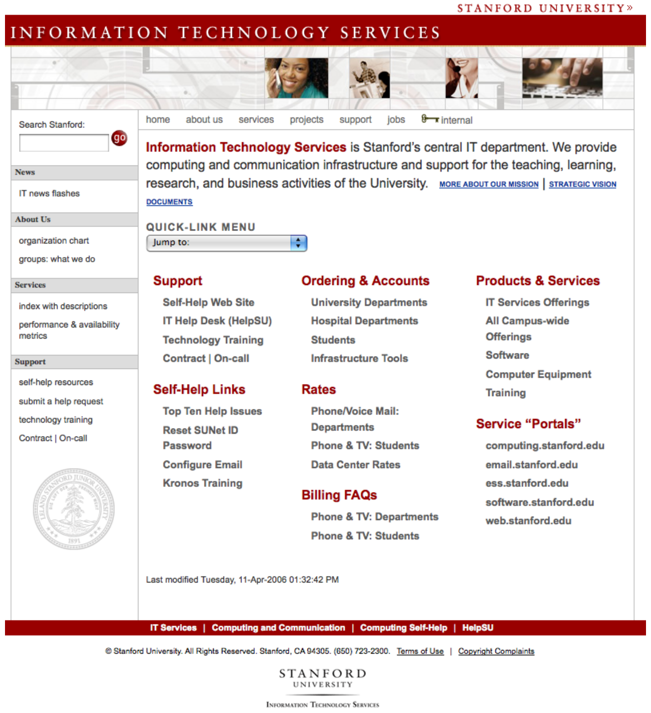
- Content: HTML 4.01
- Presentation: CSS1





Brief history… 2001-2006

- Content: XHTML 1
- Presentation: CSS2



Brief history… 2007-present

- Content: HTML5
- Presentation: CSS3

Pros and Cons of Using CSS

- Pros
  - Greater designer control of the appearance of the page
  - Easier management of site-wide changes
  - Greater accessibility to web sites by non-graphical browsers and web-page-reading software

- Cons
  - Different browsers may interpret Style Sheets in different ways
  - Some styles may not be seen at all on some browsers

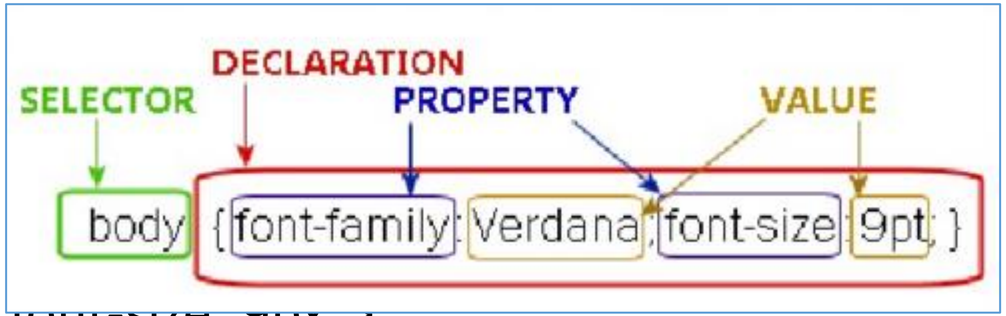
General Syntax

- Style Definition:

  Selector {

  property: value;

  }



- Example: body { font-family: Verdana; font-size: 9px; }

- The selector can either be a grouping of elements, an identifier, class, or single HTML element (body, div, etc.)
- Each declaration includes a CSS property name and a value, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces

CSS Syntax

- Case insensitive
- In most cases the use of whitespaces does not matter, within a selector it does matter specifying which elements are targeted.
- **Comments:** Comments are useful for organizing styles, annotating code, or communicating with team members.

```
/*
    This is
    a comment
*/
```

Source of Styles

- **Author / Developer Style Sheets**
  - Inline Style Sheets
  - Embedded Style Sheets
  - External Style Sheets
- **User Style Sheets**
- **Browser Default Style Sheets**

Source of Styles
> Author (Developer) Style Sheets

- **Inline Styles:**
  - Allows you to format HTML elements directly as attribute of that element itself.
  - This is achieved through the style attribute which is supported by every HTML tag.

```
<header>
  <h1 style="color:#dddddd">Wellcome to my site</h1>

 </header>
```

Wellcome to my site

  - **The simplest way but repetitive across multiple HTML elements**

Source of Styles
> Author (Developer) Style Sheets

- **Embedded Styles:**
  - Embedded styles allow you to write styles to control the page content in the head of the page's HTML.
  - This is achieved through the **style** tag.
  - Inefficient as a means of

Wellcome to my site

This is a sample site which describes how we impliment css3 into our page.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
 <!-- <meta http-equiv="refresh" content="5 ; url=home.html">-->
 <meta name="description" content="this is an introduction toi html5">
 <meta name="keywords" content="html5,intoducy,">
 <meta name="robots" content="index,follow">
<base target="_blank" href="http://localhost/html5/">
 <link rel="icon" href="assets/html5.gif">

<style>
  h1
  {
    color:#dddddd;

  }
  p
  {
    font-size: 18px;
    color:red;
  }

</style>

  <title>html 5</title>
</head>
<body>

<header>
  <h1>Wellcome to my site</h1>
<p>This is a sample site which describes how we impliment css3 into our page.
 </header>
```

Source of Styles
> Author (Developer) Style Sheets

- **External Styles**
  - Allow controlling styles across multiple pages or even an entire site.
  - This can achieved in two ways.
  - One is through the link tag (Empty tag) specified in the pages head and a separate (independent) stylesheet file.
    - **type** attribute: specifies the MIME type of the resource
    - **rel** attribute: defines the relationship that the linked resource has to the document from which it's referenced.

```css
h1
{
  color:#dddddd;

}
p
{
  font-size: 18px;
  color:red;
}
```

style.css

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
 <!-- <meta http-equiv="refresh" content="5 ; url=home.html">-->
 <meta name="description" content="this is an introduction toi html5">
 <meta name="keywords" content="html5,intoducy,">
 <meta name="robots" content="index,follow">
<base target="_blank" href="http://localhost/html5/">
 <link rel="icon" href="assets/html5.gif">
<link rel="stylesheet" type="text/css" href="style.css">



  <title>html 5</title>
</head>
<body>

<header>
  <h1>Wellcome to my site</h1>
<p>This is a sample site which describes how we impliment css3 into our page
 </header>
```

Source of Styles
> Author (Developer) Style Sheets

- **External Styles**
  - Another way to associate a stylesheet file to an HTML page is through the <style> tag and an inline style rule called @import.

style.css

```css
h1
{
  color:#dddddd;

}
p
{
  font-size: 18px;
  color:red;
}
```

```html
 <style>
@import url("style.css");

</style>
```

  - If you have to use @import, it must be specified above any rules in the style tag.

Source of Styles
> User Style Sheets

- **User Style Sheets:** This file contains the user created styles.

[firefox profile folder]/chrome/userContent.css is the current user's style sheet file for the firefox.

They provide a way for users to customize the appearance of websites according to their preferences or specific needs.

Source of Styles
> Browser Default Style Sheets(user agent style sheets)

- **Browser Default Style Sheets**: This file contains default styles for all users of a browser.

[firefox profile folder]/res/html.css is the default style sheet file for the firefox.

resource://gre-resources/forms.css

resource://gre-resources/html.css

Structuring HTML Correctly

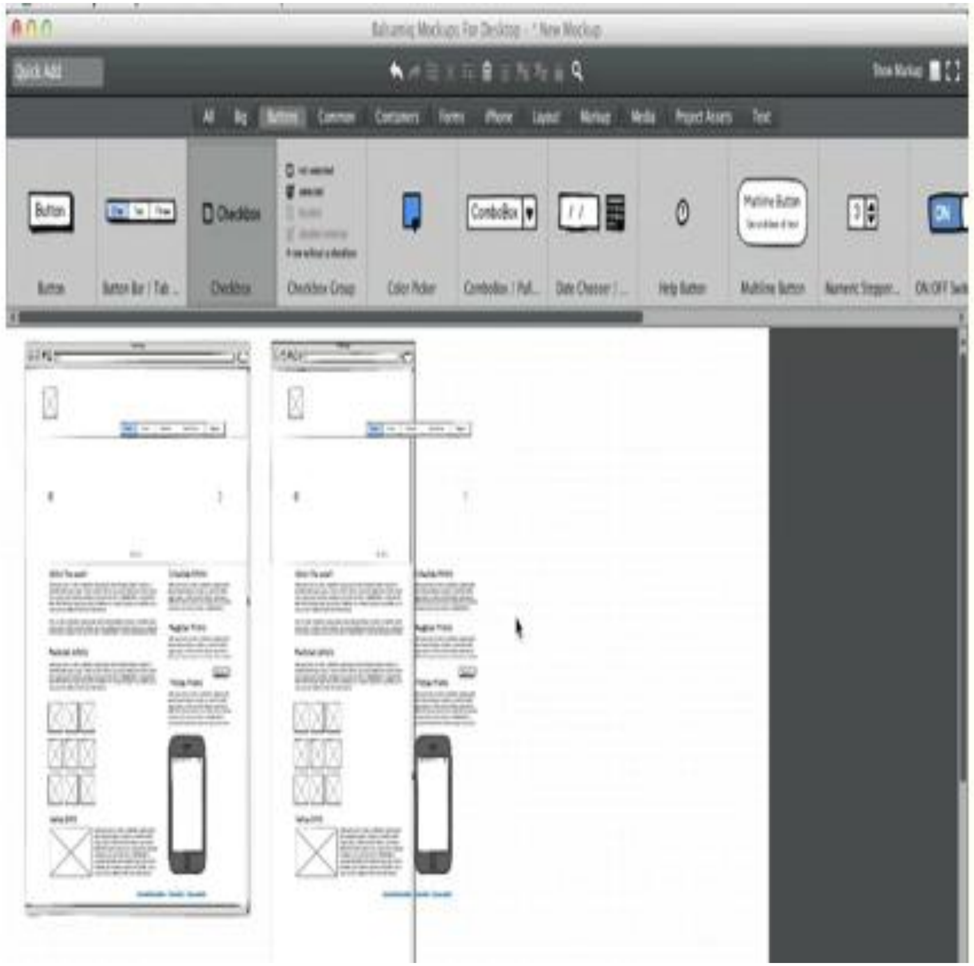
- Without a logical, consistent structure to your HTML, writing efficient CSS is impossible.
- **HTML Authoring Best Practices**
  - Focus on using **clear, semantic code.**
  - Structure your **code consistently** throughout your site.
  - Simplify your code whenever possible. **Avoid non-semantic markup.**
- **Coding Strategy**
  - Approach initial coding with only the structure and content in mind.
  - Design is considered, but should not influence code structure unless absolutely necessary.
  - This approach creates lean, semantic markup that is easier to style.
  - Focus on the relationship between HTML structure and CSS organization.

Use tools for website designing ,planning and prototyping

- Many developers and designers use good old pen and paper to plan their designs, however this doesn’t really fit well with larger projects, where collaboration / signoff is required from many different people.
- Others jump straight to HTML to create mockups – which I feel can limit creativity a little, as it puts you in the mindset of what can be done with HTML / CSS. ***Some of these prototyping tools for web design might be useful for prototyping purposes – some of them with social features.***

- http://www.webdistortion.com/2009/02/22/useful-online-tools-for-easier-website-planning-and-prototyping/

Planning and prototyping(using mindmeister and mockups)



CSS SELECTORS

```
background-color: #555;

}
h1
{
  color:#dddddd;

}
p
{
  font-size: 18px;
```

CSS Selectors

- In CSS, selectors are patterns used to select that element(s) you want to style.
- Selectors allow us to target elements on a page.
- **Selectors:** Type/Element/Tag based selector, Identifier Selector, Class Selector, Grouping Selector, Descendant Selector, Child Selector, Adjacent Sibling Selector, General Sibling Selector, Attribute Selector, Universal Selector, Pseudo-classes.

CSS Selectors
> Element Selector

- Also known as type selector and tag based selector.
- The element selector allows you to target content on the page based on the element that contains it.
- Specify the style(s) for a single HTML

```css
header
{
background-color: #555;

}
h1
{
  color:#dddddd;

}
p
{
  font-size: 18px;
  color:red;
}
```


```html
<body>

<header>
  <h1>Wellcome to my site</h1>
<p>This is a sample site which describes how we impliment css
 </header>


</body>
```

Quiz:-Use Element Selector

Css Example

This is the first pagaraph [Link one with title home]

About US

This is the second pagaraph [Link Two with title about]

This is the third pagaraph [Link Three with title contact Us]

CSS Selectors
> Class Selector (.)

```
.title
{

color: red;

}
```

```
<header>
  <h1 class="title">Wellcome to my site</h1>
<p class="title">This is a sample site which describes how we impliment css3 into ou
 </header>
```

- Class selectors allow you to target any element on the page that has the same class attribute.
- Class based styles can be used by multiple HTML elements.
- They are defined using the period (.) c
  with a class

Wellcome to my site

This is a sample site which describes how we impliment css3 into our page.

CSS Selectors
> Id Selector (Identifier Selector) (#)

- Id selectors work in much the same way as class selectors in that, they allow us to target any element on the page with a specific Id attribute.
- They are defined using the pound (#) character together with the Id of the element we want to target.
- In an event where Id and Class selector conflict with each other, the Id selector styling will be used in favor of the class because it is more specific.

```
<style>
    #aside1{
        background: beige;
    }
    #aside2{
        background: tan;
    }
</style>
```

```
<aside id="aside1">
    <h2>This is the first aside's heading</h2>
    <p>This aside discusses upcoming events</p>
</aside>
<aside id="aside2">
    <h2>This is the second aside's heading</h2>
    <p>This aside discusses featured specials</p>
</aside>
```

CSS Selectors
> Considerations (Id & Class)

```
<style>
    .bold {
        font-weight: bold;
        color: navy;
    }
</style>
```

- Class and Id attributes extend the meaning of your HTML code.

```
<ul>
    <li>Toby Malina will be in Miami on <span class="bold">January 15th.</span></li>
    <li>A new exhibit from Jeff Layton will open in New York starting in <span class="bold">May</span>.</li>
    <li>Meet Shea Hansen at the Cheek Chastain gallery in Columbia, SC on <span class="bold">April 13th</span></li>
</ul>
```

```
<style>
    .date {
        font-weight: bold;
        color: navy;
    }
</style>
```

```
<ul>
    <li>Toby Malina will be in Miami on <span class="date">January 15th.</span></li>
    <li>A new exhibit from Jeff Layton will open in New York starting in <span class="date">May</span>.</li>
    <li>Meet Shea Hansen at the Cheek Chastain gallery in Columbia, SC on <span class="date">April 13th</span></li>
</ul>
```

CSS Selectors
> Considerations (Id & Class)

```html
<aside>
    <h1>Featured Artist</h1>
    <h2>Simon Allardice</h2>
    <p>I am a paragraph that contains information pertaining to the secondary area.
    I include information about Simon Allardice as well.</p>
</aside>
```

```html
<aside id="featuredArtist">
    <h1>Featured Artist</h1>
    <h2 class="artist">Simon Allardice</h2>
    <p>I am a paragraph that contains information pertaining to the secondary area.
    I include information about <span class="artist">Simon Allardice</span> as well.</p>
</aside>
```

```html
<style>
    .artist{
        font-weight: bold;
        font-style: italic;
        color: navy;
    }
</style>
```

```html
<style>
    #featuredArtist{
        width: 300px;
        padding: 15px;
        background: tan;
    }
</style>
```

CSS Selectors
> Class and Id Selectors with the Element

- Below is an example of an element specific selector with class and id.
- Notice that there is no space between the element name and the class or id selector.

```
<style>
    .artist{
        font-weight: bold;
        font-style: italic;
        color: navy;
    }
</style>
```

```
<style>
    h2.artist {
        font-style: normal;
        color: black;
        font-variant: small-caps;
    }
</style>
```

```
<style>
    #featuredArtist{
        width: 300px;
        padding: 15px;
        background: tan;
    }
</style>
```

```
<style>
    aside#featuredArtist {
        width: 300px;
        padding: 15px;
        background: tan;
    }
</style>
```

Quiz:-use class & id selectors

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> Universal Selectors (*)

- Universal selectors are used to select any element.
- The following code sets the text color of all elements in a page to blue.

```
<style>
    * {
        color: blue;
    }
</style>
```

CSS Selectors
> Grouping Selectors (,)`

- Often you will find several different elements on the page require the exact same styling.
- Allows you to specify a single style for multiple elements at one time.
- Multiple selectors can be grouped in a single declaration by using comma (,)

```
<style>
    aside, article, section, header, footer, nav {
        display: block;
    }
</style>
```

```
<style>
    h1, h2, #featuredArtist, .artist{
        font-family: Georgia;
        font-size: 1.4em;
        font-weight: normal;
    }
</style>
```

Quiz:-Group selectors

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> Descendant Selector (space)

- The most powerful targeting ability CSS gives us is the ability to combine selectors together in what is known as descendant selectors.
  - **Allow to more precisely target content based on the relationship between nested tags and their parents.**
- Descendant selectors are used to select elements that are descendants (not necessarily children) of another element in the document tree.

```
<style>
    header h1 {
        background: #42403E;
        text-align: center;
        line-height: 75px;
        color: white;
    }
    .upcoming h2 {
        background: #7D6855;
        padding: 5px 0;
        text-align: center;
        color: white;
    }
    article aside p {
        line-height: 1.2;
        margin: 0 15px 1em;
    }
</style>
```

```
<header>
    <h1>This is the site's main heading</h1>
</header>
<article>
    <h1>This is the article's main headline</h1>
```

CSS Selectors
> Descendant Selector (space)

```
<style>
    div.abc p {
        font-weight: bold;
    }
</style>
```

```
<div class="abc">
    <div>
        <p>
            Hello there!
        </p>
    </div>
</div>
```

Quiz:-use a descendant selector to change the font size of link two

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> Child Selectors (>)

- A child selector is used to select an element that is a **direct child** of another element (parent). Child selectors will not select all descendants, **only direct children**.

```
<style>
    aside ol>li {
        font-style: italic;
        color: red;
    }
</style>
```

```
<aside class="specials">
    <ul>
        <li>This is a list of specials</li>
        <li>
            Each item is a separate special
            <ol>
                <li>point one</li>
                <li>point two</li>
                <li>point three</li>
            </ol>
        </li>
        <li>The final point</li>
    </ul>
</aside>
<h2>Subheading for upcoming dates</h2>
<ul>
    <li>upcoming date 1</li>
    <li>
        upcoming date 2
        <ol>
            <li>appearance 1</li>
            <li>appearance 2</li>
            <li>appearance 3</li>
        </ol>
    </li>
    <li>final date</li>
</ul>
```

CSS Selectors
> Child Selectors (>)

```
<style>
    div.abc > p {
        font-weight: bold;
    }
</style>
```

```
<div class="abc">
    <p>Hello there 1! (Affected)</p>
    <div>
        <p>
            Hello there 2! (Not affected)
        </p>
    </div>
</div>
```

Quiz:-Use Child Selectors to change colors of the first & third paragraph

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> Adjacent Selectors (+)

- Also called adjacent sibling selectors.
- Sibling = has the same parent
- Adjacent = immediately following
- Allow you to target elements based on which elements follow one another in your code. Essentially adjacent selector allow you to style an element based on which element comes before it in the document, **providing that both of those elements are inside the same parent.**

```
<style>
    h2 + p {
        margin-top: .6em;
        font-style: italic;
    }
</style>
```

```
<h2>This is a subheading</h2>
<p>This is more paragraph text. I'd like to do something special
with the first paragraph after a subheading.
Perhaps italicize it, or adjust the margin between them.</p>
<p>This is another paragraph, more body copy that follows the subheading
but it is not directly after it, so I don't want it to receive the
same styling as the paragraph above it.</p>
<p>Yet another body copy paragraph, this one is only here to
add weight to the styling.</p>
```

CSS Selectors
> Adjacent Selectors (+)

```
<style>
    div.abc + p {
        font-weight: bold;
    }
</style>
```

```
<div>
    <div class="abc">Message</div>
    <p>Hello there! (Affected)</p>
    <p>Hello there! (Not affected)</p>
</div>
```

Quiz:- Use Adjacent Selectors To change the last paragraph color

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> General Sibling Selectors (~)

- Sibling = has the same parent element
- General = just following
- Allow you to target elements based on which elements follow in your code. Essentially general sibling selector allow you to style an element based on which element comes before it (not necessarily immediately) in the document, providing that both of those elements are inside the same parent.

```
<style>
    h2 ~ p {
        margin-top: .6em;
        font-style: italic;
    }
</style>
```

```
<p> Will not be matched. </p>
<h2>Heading</h2>
<p>Will be matched.</p>
<div>
    <p>Will not be matched.</p>
</div>
<p>Will be matched.</p>
```

Quiz:-Use General Sibling Selectors

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> Attribute Selectors

- Attribute selectors selects elements based upon the attributes present in the HTML tags and their value.
- **Attributes are specified using brackets.**
    - ***[attribute="value"] (case sensitive)***
    - ***[attribute] (doesn't matter what the value is)***

```
<style>
    img[src="small.gif"] {
        border: 1px solid #000;
    }
</style>
```

```
<img src="small.gif" />
```

CSS Selectors
> Attribute Selectors

```
<style type="text/css">
    a[title]{
        color: red;
    }
</style>
```

```
<ul class="ul1">
    <li class="li1"><a href="http://www.zekarias.me/">This is an external link</a></li>
    <li class="li1"><a href="about.htm" title="visit about us">This is an local link</a></li>
    <li class="li1"><a href="_docs/plan.pdf" title="download my annual report">This is a link to a resource</a></li>
</ul>
```

- This is an external link
- This is an local link
- This is a link to a resource

CSS Selectors
> Attribute Selectors

```css
a[title="visit zekarias.me"]{
    color: red;
}
```

```html
<ul class="ul1">
    <li class="li1"><a href="http://www.zekarias.me/" title="visit zekarias.me">This is an external link</a></li>
    <li class="li1"><a href="about.htm" title="visit about us">This is an local link</a></li>
    <li class="li1"><a href="_docs/plan.pdf" title="download my annual report">This is a link to a resource</a></li>
</ul>
```

- This is an external link
- This is an local link
- This is a link to a resource

CSS Selectors
> Attribute Selectors

```
<style type="text/css">
    p[class=red]
    {
        color: red;
    }
</style>
```

```
<p class="red blue green">This paragraph has the classes red, blue, and green.</p>
<p class="red green">This paragraph has the classes red, and green.</p>
<p class="red blue">This paragraph has the classes red, and blue.</p>
<p class="red">This paragraph only has the class red.</p>
```

This paragraph has the classes red, blue, and green.

This paragraph has the classes red, and green.

This paragraph has the classes red, and blue.

This paragraph only has the class red.

Quiz :- Use Attribute selectors to change the color of the second link

Css Example

This is the first pagaraph Link one with title home

About US

This is the second pagaraph Link Two with title about

This is the third pagaraph Link Three with title contact Us

CSS Selectors
> Attribute Selectors

```
<style type="text/css">
    p[class~="red"]
    {
        color: red;
    }
</style>
```

- Allow us to much patterns as well
  - Tilde (~): It basically says go ahead and look for a white space separated list that includes the word.

```
<p class="red blue green">This paragraph has the classes red, blue, and green.</p>
<p class="red green">This paragraph has the classes red, and green.</p>
<p class="red blue">This paragraph has the classes red, and blue.</p>
<p class="red">This paragraph only has the class red.</p>
```

This paragraph has the classes red, blue, and green.

This paragraph has the classes red, and green.

This paragraph has the classes red, and blue.

This paragraph only has the class red.

CSS Selectors
> Attribute Selectors

```
<style type="text/css">
    a[href^="http://"]
    {
        color: red;
    }
</style>
```

- Another pattern matching character
  - Caret (^): It basically says go ahead and match any string that begins with a certain value.

```
<ul class="ul1">
    <li class="li1"><a href="http://www.zekarias.me/" title="visit zekarias.me">This is an external link</a></li>
    <li class="li1"><a href="about.htm" title="visit about us">This is an local link</a></li>
    <li class="li1"><a href="_docs/plan.pdf" title="download my annual report">This is a link to a resource</a></li>
</ul>
```

- This is an external link
- This is an local link
- This is a link to a resource

CSS Selectors
> Attribute Selectors

- Another pattern matching character
  - Dollar Sign ($): It basically says go ahead and match any string that ends with a certain value.

```
<style type="text/css">
    a[href$="pdf"]
    {
        color: red;
    }
</style>
```

```
<ul class="ul1">
    <li class="li1"><a href="http://www.zekarias.me/" title="visit zekarias.me">This is an external link</a></li>
    <li class="li1"><a href="about.htm" title="visit about us">This is an local link</a></li>
    <li class="li1"><a href="_docs/plan.pdf" title="download my annual report">This is a link to a resource</a></li>
</ul>
```

- This is an external link
- This is an local link
- This is a link to a resource

CSS Selectors
> Attribute Selectors

- Another pattern matching character
  - Asterisk Sign (*): It basically says go ahead and match any string that contains a certain value.

```
<style type="text/css">
    a[title*="visit"]
    {
        color: red;
    }
</style>
```

```
<ul class="ul1">
    <li class="li1"><a href="http://www.zekarias.me/" title="visit zekarias.me">This is an external link</a></li>
    <li class="li1"><a href="about.htm" title="please visit about us">This is an local link</a></li>
    <li class="li1"><a href="_docs/plan.pdf" title="download my annual report">This is a link to a resource</a></li>
</ul>
```

- This is an external link
- This is an local link
- This is a link to a resource

Quiz:-Use attribute selector & Asterisk Sign

Css Example

This is the first pagagraph Link one with title home

About US

This is the second pagagraph Link Two with title about

This is the third pagagraph Link Three with title contact Us